جامعة الملك فهد للبترول والمعادن
King Fahd University of Petroleum & Minerals

College of Computing and Mathematics
Information and Computer Science Department

ICS 202: Data Structures and Algorithms (3-3-4)

Syllabus- Second Semester 2026-2027 (261)

Class Time, Venue and Instructor Information:

Section	Time	Venue	Instructor	Office Hours
01	UTR 08:00 – 08:50	24 -236-2	Moayad Alnammi Office: 22-322 Phone: 013-860-1215 E-mail: moayad@kfupm.edu.sa	UTR: 10:00 – 11:00 or by appointment
02	UTR 09:00 – 09:50	22-127		
05	UTR 11:00 – 11:50	22-132		
03	UTR 08:00 – 08:50	24 -236-2	Qasim Umer Office: 22-218 Phone: 013-860-5391 E-mail: qasim.umer@kfupm.edu.sa	UTR: 13:00 – 14:00 or by appointment
06	UTR 12:00 – 12:50	24-180		
04	UTR 08:00 – 9:50	22 - 132	Alwaleed Alkhodairi Office: 22-115-1 E-mail:a.alkhodairi@kfupm.edu.sa	UTR: 11:00 – 12:00 or by appointment
F11	UTR 08:00 – 08:50	24 - 158	Rashad Othman Email: rashad.othman@kfupm.edu.sa Tel: 013-860-8839 Office: 22-106	UTR: 10:00 – 11:00 UT: 13:00 – 14:00 or by appointment
F12	UTR 09:00 – 09:50	24 - 128		
F13	UTR 11:00 – 11:50	59 - 1003		
F14	UTR 12:00 – 12:50	59 - 1009		

Course Description

Review of object-oriented concepts; Basic algorithms analysis; Fundamental data structures - implementation strategies for stacks, queues and linked lists; Recursion; Implementation strategies for tree and graph algorithms; Greedy Algorithms; Hash tables; Applications of data structures (e.g., data compression and string matching).

Prerequisite: ICS 108

Course Objectives

The objectives of this course are to:

1. Introduce students to fundamental data structures, their algorithms, implementations and applications.
2. Teach students how to analyze the efficiency of the fundamental data structures in terms of both time and space so that they can decide what data structure is suitable for a given problem

Course Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Apply object-oriented concepts (inheritance, polymorphism, etc.) in software design. [SO 2]
2. Implement various data structures and their algorithms and apply them in implementing simple applications. [SO 2]
3. Analyze simple algorithms and determine their efficiency using big-O notation. [SO 1]
4. Analyze the efficiency of the fundamental data structures in terms of both time and space. [SO 1]
5. Decide which data structures are suitable for which applications. [SO 2]
6. Apply the knowledge of data structures to other application domains such as data compression and hashing. [SO 2]

Textbooks

1. Adam Drozdek, "Data Structures and Algorithms in Java", Thomson Learning, 4th edition, 2013, ISBN 978-981-4239-23-3.

Tentative Weekly Schedule

Week #	Date	Lecture Topic	Lab Topic	Tentative deadlines
W #1	20-Aug	Overview	No Lab	
W #2	23-Aug	Complexity Analysis	No Lab	
	25-Aug	Complexity Analysis		
	27-Aug	Complexity Analysis		
W #3	30-Aug	Complexity Analysis	Complexity Analysis	HW #1 is posted
	1-Sep	Singly Linked Lists		
	3-Sep	Singly Linked Lists		
W #4	6-Sep	Doubly Linked Lists	Singly Linked Lists	HW #1 is due
	8-Sep	Doubly Linked Lists		
	10-Sep	Stacks		
W #5	13-Sep	Stacks	Doubly Linked Lists	HW #2 is posted
	15-Sep	Queues		
	17-Sep	Recursion		
W #6	20-Sep	Recursion	Stacks and Queues	
	22-Sep	Recursion & Recursion Analysis		
	24-Dep	**Saudi National Day Holiday**		
W #7	27-Sep	Recursion Analysis	Recursion	HW #2 is due
	29-Sep	Binary Trees		
	1-Oct	Binary Trees		
W # 8	4-Oct	Binary Search Trees	Lab Quiz 01	HW #3 is posted
	6-Oct	Binary Search Trees		
	7-Oct	**Midterm Exam**		
	8-Oct	AVL Trees		
W #9	11-Oct	AVL Trees	Binary and Binary Search Trees	
	13-Oct	AVL Trees		
	15-Oct	Binary Heaps		
W #10	18-Oct	Binary Heaps	No Lab	HW #3 is due
	20-Oct	**Midterm Break**		
	22-Oct	**Midterm Break**		
W #11	25-Oct	B Trees	Binary Search and AVL Trees	HW #4 is posted
	27-Oct	B Trees		
	29-Oct	B Trees		
W #12	1-Nov	Hashing	Binary Heaps	
	3- Nov	Hashing		
	5- Nov	Hashing		
W #13	8- Nov	Data Compression	Hashing	
	10-Nov	Data Compression		
	12-Nov	Data Compression		
W #14	15-Nov	Graph Representations and Traversals	Data Compression	HW #4 is due
	17-Nov	Graph Representations and Traversals		
	19-Nov	Graph Representations and Traversals		
W #15	22-Nov	**Autumn break**	No Lab	
	24-Nov	Graph Representations and Traversals		
	26-Nov	Graph and Greedy Algorithms		
W #16	29-Nov	Graph and Greedy Algorithms	Graph Representations and Traversals	
	1-Dec	Graph and Greedy Algorithms		
	3-Dec	Graph and Greedy Algorithms		
W#17	6-Dec	String Matching	Lab Quiz 02	
	8-Dec	String Matching		
	10-Dec	String Matching		

Assessment Plan

Assessment Tool	Details	Weight
Homework	3 up to 4 assignments	5%
Class Work	**Quizzes** and **activities**	20%
Lab	Lab Tasks: (5%) Lab project: (5%) Two lab Tests: (5% each)	20%
Midterm Exam: TBA	TBA	25%
Final Exam: TBA	TBA	30%
Total		100%

Course policy

- ***Course Website & Participation***: All course material will be made available on Blackboard. Students are required to periodically check Blackboard and download course material as needed.
- ***Attendance***: Regular attendance is a university requirement; therefore, attendance will be checked in each class. Missing more than 9 lectures and 3 labs will result in a DN grade without prior warning.
- ***Projects and Homework***: Homework assignments (according to the instructions given with each assignment) are to be submitted, through Gradescope, by the due date and time indicated. **No late homework will be accepted**. Discussion with classmates is encouraged, but **COPYING** solutions is strictly prohibited and considered **CHEATING**.
- ***No makeup of homework assignments, quizzes or exams will be given***.
- ***Re-grading policy***: If you have a complaint about any of your grades, discuss it with the instructor no later than three days after the day grades were distributed.
- ***Office Hours***: Students are encouraged to use office hours to clarify any part of the material that is not clear and discuss the project. If it is not suitable for you, please do not hesitate to e-mail me to set an appointment.
- ***Academic honesty***: Students are expected to comply with all the university regulations on academic honesty. Cheating will be reported to the Department Chairman and will be severely penalized.